

Polynomial rings

Observation

(Homomorphism property
for polynomial rings)

$$\left\{ \varphi: \mathbb{Z}[x] \xrightarrow[\text{homomorphism}]{\text{ring}} R' \right\}$$

φ

$\downarrow$

$\varphi(x)$

injection

$\downarrow$

$\{ r \in R' \}$

$(\sum a_i x^i \mapsto \sum a_i r^i): ev_r$

$\downarrow$

"evaluation at r "

All $Z[x] \rightarrow R'$
are evaluation homomorphisms

R : Commutative ring

$$\{ \varphi: R[x] \longrightarrow R' \}$$

bijection

$$\left\{ (f, r) : \begin{array}{l} \text{Evaluating constant} \\ \text{polynomials} \\ f: R \rightarrow R' \text{ ring hom.} \\ * \\ r \in R' \\ \text{Evaluating } x \end{array} \right\}$$

$$ev_{(f, r)} : \left(\sum a_i x^i \right) \mapsto \left(\sum f(a_i) r^i \right)$$

$\uparrow$ $\uparrow$
 $R[x]$ R'

$$1_R \mapsto f(1) = 1_{R'}$$

$\uparrow$
 $1_{R[x]}$

Rem:

$$\left\{ \begin{array}{l} \text{Group homomorphism} \\ f: \mathbb{Z} \longrightarrow G \\ \quad \quad \quad (m \mapsto g^m) \end{array} \right\}$$

$\downarrow$ $\downarrow$

$f(1)$ g

$\{ g \in G \}$

Polynomials over fields

k : field

Observation (2)

Every non-zero polynomial
in $k[x]$ is monic

Observation (3) (Everything is in $k[x]$)

If $f(x)$ is irreducible

& $f(x) \nmid a(x)$, then $\left[\begin{array}{l} \deg \geq 1 + \\ f(x) \nmid a(x)c(x) \\ \deg c(x), \deg c(x) \\ < \deg f(x) \end{array} \right]$

$\exists s(x), t(x) \in k[x]$ s.t.

$$1 = s(x)f(x) + t(x)a(x).$$

(Bezout!)

Pf:

$$a(x) = q_1(x)f(x) + r_1(x)$$

$$f(x) = q_2(x)r_1(x) + r_2(x)$$

$$r_1(x) = q_3(x)r_2(x) + r_3(x)$$

$$r_{n-1}(x) = q_{n+1}(x)r_n(x) + \underline{r_{n+1}(x)}$$

$$r_n(x) = q_{n+2}(x)r_{n+1}(x)$$

$$\begin{aligned} r_{n+1}(x) \\ = q_{n+1}(x)r_n(x) \\ - r_{n-1}(x) \end{aligned}$$

$$\deg f(x) > \deg r_1(x) > \deg r_2(x) > \dots$$

$$\Rightarrow \sqrt{r_{n+1}(x) \mid f(x)} \Rightarrow r_{n+1}(x) \in k^*$$

$$\Sigma) I = U^{-1} U = \underbrace{U^{-1} S(x)}_{s(x)} \underbrace{U t(x)}_{t(x)} + f(x)$$

rename.

Rmk: (1) The proof only used that $f(x)$ & $a(x)$ have no non-constant common factors

(2) In general, the end guy $r_{n+1}(x)$ is "the" greatest common divisor of $f(x)$ & $a(x)$

i.e. $r_{n+1}(x)$ is the polynomial with highest degree dividing

both $f(x)$ & $a(x)$

$$r_{n+1}(x) = \gcd(f(x), a(x))$$

(General Bezout)

$$r_{n+1}(x) = s(x)f(x) + t(x)a(x) \\ \exists s(x), t(x) \in k[x]$$

Constructing new fields from old fields

k : field

Problem 7 : $f(x) \in k[x]$ irred.

$\Rightarrow k[x]/(f(x))$ is a field

Observation (4)

$\varphi: k \longrightarrow R'$ ring homomorphism

$\ker \varphi = \{0\} \Leftrightarrow \varphi$ is injective.

Pf: $a \in \ker \psi$

$$\psi(a) = 0$$

$$\text{If } a \neq 0 \Rightarrow a^{-1} \cdot a = 1$$

$$\Rightarrow \psi(a^{-1} \cdot a) = \psi(1) = 1$$

$$\parallel \quad \psi(a^{-1}) \cdot \psi(a) = 0 \quad \text{X}$$

Observation (5)

(Homomorphism property for $k[x]/(f(x))$)

$$f(x) = \sum a_i x^i$$

Suppose that we have

a ring homomorphism $\lambda: k \rightarrow B$

Then

$$\left\{ \begin{array}{c} k[x]/(f(x)) \xrightarrow{\quad} B \\ \uparrow \lambda \\ k \end{array} \right\} \longleftrightarrow \left\{ h \in B : \begin{array}{l} \text{"}f(h)\text{"} = \sum_i \lambda(a_i) h^i = 0 \end{array} \right\}$$

Pls

הערה

$$\ell V_{2,1} : \Sigma \mathbb{A}_i x^i \mapsto \Sigma \mathbb{A}_i v^i$$
$$\psi: \psi(f(h)) = 0$$
$$\mathbb{R}[x] / (f(x))$$

injective

$$\rightarrow \uparrow$$

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B $\Rightarrow$ h

$$k \longrightarrow k[x] \longrightarrow k[x]/(f(x))$$
$$a \Vdash a \Vdash a \neq (f(x))$$